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Executive Summary

In choosing high-priority research topics across technology, health, environment, economics, and geopolitics, we focus on globally urgent issues. The five proposed topics are **1) AI Governance and Ethics**, **2) Global Health Security and Pandemic Preparedness**, **3) Climate Change Mitigation and Adaptation**, **4) Economic Resilience (Inflation, Inequality, Supply Chains)**, and **5) Geopolitical Stability (Great Power Competition)**. Each is supported by authoritative sources and current trends. We identify *Climate Change Mitigation and Adaptation* as the most globally pressing topic. The subsequent analysis section provides an executive summary, detailed evidence, and recommendations for this topic. Finally, we outline a 2-week research plan with milestones. The report includes tables comparing topics and their key sources, as well as Mermaid diagrams for the research timeline and inter-domain relationships.

Proposed Research Topics

1. AI Governance and Ethics (Technology)

Rapid advances in artificial intelligence (AI) promise significant economic and social benefits but also pose risks (misinformation, bias, privacy) that demand robust governance. AI is borderless, requiring international coordination. We focus on policy frameworks (OECD AI Principles, UNESCO ethics standards) and ethical norms to balance innovation with safety【26†L3865-L3873】【20†L75-L82】.

- **Key Questions:** What global AI governance frameworks exist and how effective are they? How do countries' AI regulations compare? What ethical principles (transparency, fairness, privacy) dominate AI guidelines? What mechanisms ensure AI safety (testing standards, liability rules)? How can innovation and security be balanced?
- **Key Sources:** UNESCO's *Recommendation on the Ethics of AI* (2021)【20†L75-L82】; OECD AI Principles (2019, updated 2024)【26†L3865-L3873】; WEF *Global Risks Report* (AI risks)【3†L103-L106】; EU AI Act drafts; World Bank's *Global Trends in AI Governance* (2023).
- **Search Keywords:** "global AI governance frameworks", "AI ethics principles UNESCO OECD", "AI regulation comparative study", "AI transparency fairness policy", "OECD AI principles implementation".

2. Global Health Security and Pandemic Preparedness (Health)

The COVID-19 pandemic exposed gaps in disease surveillance, healthcare capacity, and international cooperation【30†L42-L50】【35†L405-L414】. Strengthening pandemic preparedness is crucial in our volatile world. We examine legal frameworks (WHO's IHR), funding, and equity in response capabilities【34†L97-L105】【35†L405-L414】.

- **Key Questions:** How effective are existing instruments like WHO's International Health Regulations (2005)【34†L97-L105】 and the proposed pandemic treaty? What financing gaps remain for global health security? How can surveillance (One Health approach) be improved for zoonotic threats? How

to rebuild trust and cooperation after COVID-19? What role do equity and capacity-building play in resilience?
- **Key Sources:** WHO – International Health Regulations overview【34†L97-L105】; Global Preparedness Monitoring Board (GPMB) *2025 Report*【30†L42-L50】; Lancet editorial on IHR reform【35†L405-L414】; WHO pandemic treaty negotiations; CDC/World Bank health security financing reports.
- **Search Keywords:** "pandemic preparedness global health security", "International Health Regulations reform", "One Health disease surveillance", "GPMB 2025 report pandemic", "global health emergency financing gap".

3. Climate Change Mitigation and Adaptation (Environment)

Climate change remains a foremost long-term global risk. The IPCC warns that human activity has already caused ~1.1°C of warming【45†L13-L20】 and current policies portend ~3.2°C by 2100【45†L126-L129】 unless dramatically tightened. Meanwhile, adaptation funding is woefully short: developing countries need ~$310–365 billion annually, but receive only ~$26 billion (2023)【58†L75-L83】 – a 12–14× gap. Research must cover reducing emissions and building resilience.

- **Key Questions:** How close are current emissions trajectories to Paris targets? What are the most effective mitigation strategies (energy transition, land use)? What is the scale of the adaptation finance gap (e.g. UNEP Adaptation Gap Report 2025)【58†L75-L83】, and how to close it? How are Nationally Determined Contributions (NDCs) evolving? What technological and policy innovations (carbon capture, climate-smart agriculture) can help?
- **Key Sources:** IPCC AR6 Synthesis Report (2023) on mitigation pathways【45†L13-L20】【45†L126-L129】; UNEP *Adaptation Gap Report 2025*【58†L75-L83】; UNFCCC NDC Synthesis Reports; WRI climate data; IEA energy scenarios; NOAA/NASA climate trends.
- **Search Keywords:** "IPCC climate change mitigation adaptation", "adaptation finance gap statistics 2025", "NDC progress UNFCCC 2025", "climate resilience strategies developing countries", "carbon emission trajectories 2030".

【79†embed_image】 *Figure: Global CO_2 emissions from fossil fuels have risen sharply since the mid-20th century【78†L73-L80】, underscoring the urgency of mitigation.*

4. Economic Resilience: Inflation, Inequality, and Supply Chains (Economics)

Global economic growth is slowing and inflationary pressures vary across regions【50†L320-L328】【83†L74-L81】. Research focuses on maintaining stability amid trade fragmentation and debt burdens. Key issues include managing inflation without recession, addressing inequality, ensuring supply chain resilience, and assessing the impact of technology on labor markets.

- **Key Questions:** What are the drivers of current inflation and debt crises? How can monetary/fiscal policy restore confidence (as IMF recommends)【50†L320-L328】? What strategies can reduce inequality and rebuild fiscal buffers? How have supply chains shifted (post-pandemic reshoring vs. diversification)? What is the economic impact of technology (AI, automation) on jobs and growth?
- **Key Sources:** IMF World Economic Outlook (Oct 2025)【50†L320-L328】; World Bank Global Economic Prospects (2026)【83†L74-L81】; OECD Economic Outlook; UNDP/MPI poverty reports; major banks' global scenario analyses; think-tank studies on supply chains.

- **Search Keywords:** “global economic growth forecast 2026 IMF”, “inflation drivers post-pandemic”, “supply chain resilience strategy”, “global inequality trends 2025”, “AI impact on labor market”.

5. Multipolar Geopolitics and Global Security (Geopolitics)

International order is fragmenting amid great-power competition. War and tension (Ukraine, Middle East) threaten stability and resource flows【13†L64-L70】【3†L173-L181】. We explore how geopolitical shifts affect security and cooperation. Key areas include power transitions (U.S.-China-Russia), cyber warfare, resource conflicts, and the future of international institutions.

- **Key Questions:** How are US-China relations evolving and what are implications for global trade and security【13†L64-L70】? What scenarios might unfold in ongoing conflicts (e.g. Ukraine, Iran-Israel)? How is the rise of regional powers reshaping alliances (e.g. India, ASEAN)? What are emerging risks in cyber/AI warfare? How do environmental and energy issues factor into security (climate wars, energy independence)?
- **Key Sources:** WEF *Global Risks Report 2025* (state-based conflict as top risk, 64% foresee power-competition)【3†L77-L85】【3†L173-L181】; S&P Global *Top Geopolitical Risks 2025* (Russia-Ukraine, US-China, cyber)【13†L64-L70】; Brookings/CFR reports on US-China tech war; NATO/UN security briefings.
- **Search Keywords:** “geopolitical risk US China conflict 2025”, “global security risks WEF”, “cyber warfare threats 2024-25”, “resource wars climate change migration”, “UN Security Council analysis”.

Top-Selected Topic: Climate Change Mitigation and Adaptation

Given its enduring global impact and urgent timeframe, **Climate Change Mitigation and Adaptation** is chosen for deep analysis. The following sections summarize evidence and recommendations for this topic.

Executive Summary

Climate change poses an existential threat: human-driven warming has reached ~1.1 °C above pre-industrial levels, and without stronger policies the world is on track to exceed 3 °C by 2100【45†L126-L129】. While nearly all nations have climate plans, emission-cutting pledges and adaptation funding remain insufficient. Developing countries need ~$310–365 billion per year by 2035 for climate adaptation, yet only $26 billion was provided in 2023【58†L75-L83】. This “adaptation finance gap” – roughly 12–14× shortfall – imperils lives and economies. To address these issues, global action should focus on (1) sharply increasing mitigation efforts (decarbonization, clean energy investment), (2) expanding and accelerating adaptation funding and infrastructure, (3) enhancing international cooperation and climate finance commitments, and (4) investing in innovation (renewables, carbon capture, resilient agriculture). Uncertainties include future climate feedbacks, political will, and economic constraints. Key recommendations include doubling adaptation finance by 2030, implementing carbon pricing or regulations, integrating climate risk into financial systems, and supporting vulnerable communities through technology transfer and capacity building.

Detailed Findings

- **Emissions Trends and Projections:** Global fossil CO_2 emissions have more than doubled since 1990 and surpassed 35 billion tonnes/year【78†L73-L80】. Per the IPCC, without additional policies, projections indicate ~3.2 °C warming by 2100【45†L126-L129】. Many countries have pledged net-zero targets (mid-century), but existing national policies fall short【45†L126-L129】. Methane and other GHG reductions are also inadequate.

- **Mitigation Measures:** Progress has been made in renewable energy deployment and energy efficiency, but not enough to curb emissions. According to the IPCC, deep near-term cuts (45% by 2030) are needed for 1.5 °C goals【45†L126-L129】. Carbon pricing exists in ~25% of emissions, but at low rates. Technological innovations (e.g., battery storage, green hydrogen) are emerging but require scale-up. Deforestation has slowed slightly due to policies, yet land-use emissions remain substantial.

- **Adaptation Gaps:** UNEP reports that developing countries' adaptation needs are $310–365 bn/yr (2035), whereas only $26 bn was delivered in 2023【58†L75-L83】. This "12–14× gap" means plans (87% of countries have an NAP/strategy) lack implementation funds【58†L75-L83】. Key vulnerabilities include coastal flooding, extreme weather, and food insecurity. Progress: more adaptation projects were implemented (1,600 reported actions)【39†L179-L185】, but financing lags.

- **Policy and Finance:** International climate finance (UNFCCC data) is rising but still far below needs. UNEP warns developed countries will miss the 2025 goal of doubling adaptation finance【56†L69-L77】. Private sector contributions are rising but require policy support (e.g. blended finance). Climate risk disclosure is increasing, yet many financial institutions do not fully account for climate-related liabilities.

- **Socioeconomic Impacts:** Climate impacts have immediate health and economic effects (heatwaves, disasters). Vulnerable nations are disproportionately affected, exacerbating poverty and inequality. Unless adaptation and resilience-building accelerate, damages will undermine development goals.

Evidence Synthesis

Multiple sources underscore the urgency: **IPCC 2023** confirms that warming has reached record levels, with current emissions trajectories far exceeding targets【45†L126-L129】. The **UNEP Adaptation Gap Report (2025)** provides hard figures on the funding shortfall【58†L75-L83】. Climate models and observational data (e.g., Berkeley Earth) indicate 2023 was the warmest year on record (≥1.5 °C)【67†L103-L111】, suggesting frequent overshoots of Paris goals.

There is broad agreement on the science (warming trends, impacts), and wide divergence in perspectives on solutions: economists stress market instruments, activists demand rapid decarbonization, and policymakers highlight technological paths. Weighing evidence: models show that delaying mitigation drastically raises future costs, while adaptation has diminishing returns if warming is uncontrolled. Hence, both mitigation and adaptation are essential and interlinked.

Interdisciplinary research (climate science, economics, engineering, social policy) converges on the need for a two-pronged approach: reduce emissions aggressively *and* strengthen resilience of communities. For example, NDC analyses reveal that ambitious National Adaptation Plans often lack funding commitments【58†L75-L83】. Studies on climate finance agree that public and private flows must increase vastly.

Uncertainties and Limitations

- **Climate Projections:** Climate sensitivity (temperature response to CO_2) remains uncertain. Feedbacks (permafrost melt, Amazon dieback) could accelerate warming beyond model estimates.
- **Policy Implementation:** Political will is uncertain; major emitters' policies can change with elections. International cooperation (e.g. on finance) is volatile.
- **Technological Change:** Breakthroughs (fusion, new carbon tech) are possible but unproven at scale; costs could drop faster or slower than expected.
- **Economic Factors:** Energy prices and economic growth will affect emissions and funding. The war in Ukraine and Middle East shows how geopolitics can disrupt climate action (e.g. shifting to coal temporarily).
- **Data Gaps:** Climate finance tracking is incomplete. Many adaptation needs are poorly quantified, especially in low-income regions.

Actionable Recommendations

1. **Scale up Climate Finance:** Developed countries should dramatically increase funding to vulnerable nations. Specifically, governments and multilateral funds must **double adaptation financing by 2030** (as per Glasgow Pact goals). This requires at least ~\$50 bn/yr mobilized by 2030, combining public, MDB, and leveraged private sources【39†L163-L170】【58†L75-L83】. Innovative instruments (debt-for-climate swaps, green bonds) should be expanded.

2. **Strengthen Emission Reduction Policies:** Major emitters must adopt more stringent policies. For example, implement or raise carbon pricing to \$100+/t CO_2 by 2030, and mandate renewables for new power generation. Rapid phase-out of coal (as per Carbon Brief: new coal plants peaked in 2025【63†L120-L124】) and oil subsidies is essential. Countries should update NDCs to close the gap between pledges and pathways to 1.5 °C【45†L126-L129】.

3. **Invest in Climate Resilient Infrastructure:** Governments and donors should prioritize resilient infrastructure (flood defenses, heat-resistant crops, water storage). This includes integrating climate risk into building codes and urban planning. Public-private partnerships can finance resilient energy grids and water systems, particularly in climate-vulnerable regions.

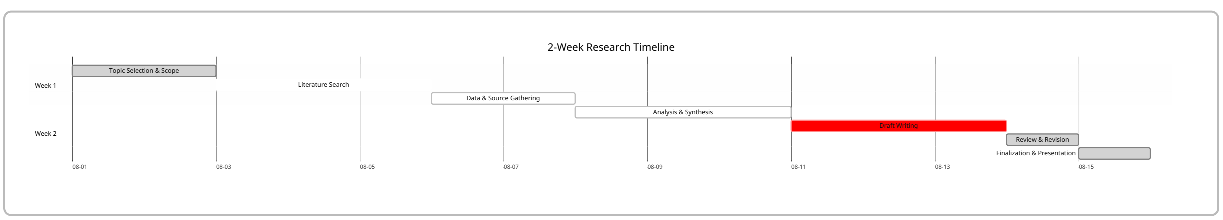
4. **Enhance Monitoring and Research:** Expand data collection on climate impacts and finance. Support global observatories (e.g., satellite monitoring of emissions, international databases of adaptation projects) and research on geoengineering risks. Evidence-based policymaking requires better metrics for adaptation outcomes and progress tracking.

5. **Promote International Cooperation:** Reinforce climate diplomacy (UNFCCC, G20) to rebuild trust. Developed countries should honor and exceed finance pledges (e.g. New Collective Quantified Goal), while emerging economies can share technology (e.g. patents for renewables). Encourage joint R&D (e.g. Mission Innovation) and capacity-building programs to disseminate clean technologies globally.

Comparative Overview of Topics

Topic	Domain	Key Sources	Example Keywords/ Queries
AI Governance & Ethics	Technology	UNESCO AI Ethics (2021)【20†L75-L82】; OECD AI Principles【26†L3865-L3873】; WEF Global Risks【3†L103-L106】; EU AI Act	“global AI governance”, “AI ethics UNESCO OECD”
Global Health Security	Health	WHO IHR summary【34†L97-L105】; GPMB 2025 report【30†L42-L50】; Lancet (2022) on IHR【35†L405-L414】; WHO pandemic treaty news	“pandemic preparedness”, “international health regulations”
Climate Mitigation & Adaptation	Environment	IPCC AR6 Synthesis (2023)【45†L13-L20】【45†L126-L129】; UNEP Adaptation Gap 2025【58†L75-L83】; UNFCCC NDC reports; IEA clean energy	“climate adaptation finance gap”, “emission reduction pathways”
Economic Resilience (inflation, supply chains)	Economics	IMF WEO Oct 2025【50†L320-L328】; World Bank Global Prospects 2026【83†L74-L81】; OECD Economic Outlook; WEF Global Risks【3†L77-L85】	“global growth forecast 2026”, “supply chain risk analysis”
Geopolitical Stability	Geopolitics	WEF Global Risks 2025 (state conflict)【3†L77-L85】【3†L173-L181】; S&P Top Geopolitical Risks【13†L64-L70】; NATO/UN security briefings	“US-China relations 2026”, “global conflict risk”

Research Plan Timeline (2 Weeks)



Milestones: selecting final topic (Day 2); completing literature review and data collection (Day 5); analysis synthesis (Day 8); drafting report (Day 11); revisions and final submission (Day 14).

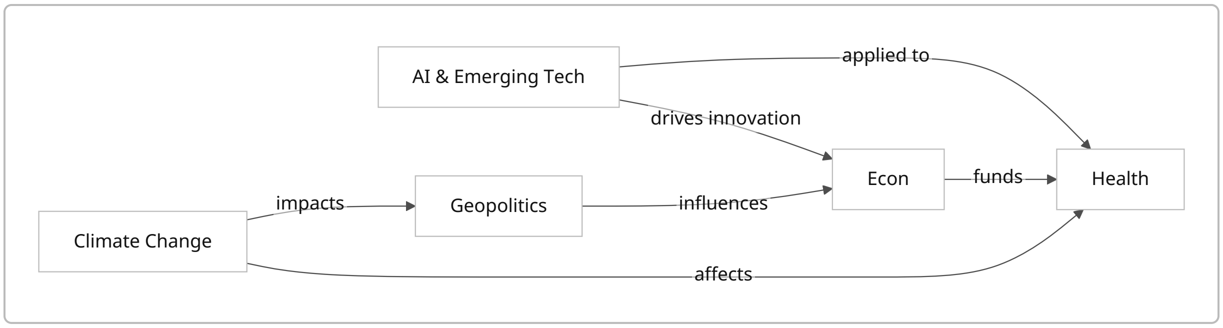


Figure: Conceptual links between domains (AI impacts economy and health; climate affects health and geopolitics; geopolitics influences economy and resources; economies fund health systems).
